

理论作业

Exercise 12.42

只需证明对 $\forall U \in L^1(h\mathbb{Z})$, 傅里叶 + 逆 = 恒等变换.

$$\text{即 } U(m) = \frac{1}{\sqrt{2\pi}} \int_{-\frac{\pi}{h}}^{\frac{\pi}{h}} e^{imh\xi} \hat{U}(\xi) d\xi$$

$$\text{因为 } U \in L^1(hF) \Rightarrow \sum_{m \in \mathbb{Z}} |U(m)| < \infty$$

$$\text{又因为 } \forall m \in F, \forall \xi \in [-\frac{\pi}{h}, \frac{\pi}{h}], |e^{-imh\xi}| \leq |U_m|.$$

$$\text{因此, } \forall \xi \in [-\frac{\pi}{h}, \frac{\pi}{h}], \frac{1}{\sqrt{2\pi}} \sum_{m \in F} |e^{-imh\xi} U_m h| \leq \frac{1}{\sqrt{2\pi}} \sum_{m \in F} |U_m| h \leq \infty$$

所以 $\hat{U}(\xi) = \frac{1}{\sqrt{2\pi}} \sum_{m \in F} e^{-imh\xi} U_m h$ 在 $\xi \in [-\frac{\pi}{h}, \frac{\pi}{h}]$ 上一致收敛, 则求和和积分可换序.

$$\begin{aligned} \hat{U}(\xi) &= \frac{1}{\sqrt{2\pi}} \sum_{m \in F} e^{-imh\xi} U(m) h \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\frac{\pi}{h}}^{\frac{\pi}{h}} \left(\frac{1}{\sqrt{2\pi}} e^{inh\xi} \sum_{n \in F} e^{-imh\xi} U(n) h \right) d\xi \\ &= \frac{1}{2\pi} \sum_{n \in F} \int_{-\frac{\pi}{h}}^{\frac{\pi}{h}} e^{i(m-n)h\xi} U(n) h d\xi \end{aligned} \quad (1)$$

$$\text{因为 } \int_{-\frac{\pi}{h}}^{\frac{\pi}{h}} e^{i(m-n)h\xi} d\xi = \begin{cases} \frac{2\pi}{h} & \text{if } m = n, \\ 0 & \text{otherwise.} \end{cases},$$

代入式(1)得证.

Exercise 12.49

θ method :

$$-\theta r U_{i-1}^{n+1} + (1 + 2\theta r) U_i^{n+1} - \theta r U_i^{n+1} = (1 - \theta) r U_{i-1}^n + (1 - 2(1 - \theta)r) U_i^n + (1 - \theta) r U_{i+1}^n$$

使用 fourier 变换:

$$\begin{aligned} U_j^{n+1} &= \frac{1}{\sqrt{2\pi}} \int_{-\frac{\pi}{h}}^{\frac{\pi}{h}} (-\theta r e^{i(j-1)h\xi} + (1 + 2\theta r) e^{ijh\xi} - \theta r e^{i(j+1)h\xi}) \hat{U}^n + 1_j(\xi) d\xi \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\frac{\pi}{h}}^{\frac{\pi}{h}} ((1 - \theta) r e^{i(j-1)h\xi} + (1 - 2(1 - \theta)r) e^{ijh\xi} + (1 - \theta) r e^{i(j+1)h\xi}) \hat{U}_j^n(\xi) d\xi \end{aligned} \quad (2)$$

(2)两边同除 $e^{ijh\xi}$ 可得:

$$(-\theta r e^{-ih\xi} + (1 + 2\theta r) - \theta r e^{ih\xi}) \hat{U}^{n+1}(\xi) = ((1 - \theta) r e^{-ih\xi} + (1 - 2(1 - \theta)r) + (1 - \theta) r e^{ih\xi}) \hat{U}^n(\xi)$$

即

$$\hat{U}^{n+1}(\xi) = \frac{1 - 4(1 - \theta)r \sin^2(\frac{h\xi}{2})}{1 + 4\theta r \sin^2(\frac{h\xi}{2})} \hat{U}^n(\xi) \quad (3)$$

要使 $|\frac{1 - 4(1 - \theta)r \sin^2(\frac{h\xi}{2})}{1 + 4\theta r \sin^2(\frac{h\xi}{2})}| < 1$, 需要 $\forall \xi \in [-\frac{\pi}{h}, \frac{\pi}{h}]$

$$-(1 - \theta) \sin^2(\frac{h\xi}{2}) < \theta \sin^2(\frac{h\xi}{2}) \quad (4)$$

$$2(1 - 2\theta)r \sin^2(\frac{h\xi}{2}) \leq 1 \quad (5)$$

因为 $\theta \in [0, 1]$, 所以(4)一定成立, 考虑(5),

当 $\theta \in [\frac{1}{2}, 1]$ 时, $2(1 - 2\theta)r \sin^2(\frac{h\xi}{2}) \leq 0 < 1$,

当 $\theta \in [0, \frac{1}{2})$ 时, $2(1 - 2\theta)r \sin^2(\frac{h\xi}{2}) \leq 2(1 - 2\theta)r < 1$, $\therefore r = \frac{kv}{h^2} \Rightarrow < \frac{h^2}{2(1 - 2\theta)v}$

Fourier 变换让我们能从频率的角度来看待问题, 将复杂的 PDE 简化为代数问题.

Exercise 12.78

$$\mu = \frac{ak}{h}$$

(12.89a)

$$\tau(x, t) = u(x, t+k) - u(x, t) + \frac{ak}{2h}(3u(x, t) - 4u(x-h, t) + u(x-2h, t)) - \frac{a^2k^2}{2h^2}(u(x, t) - 2u(x-h, t) + u(x-2h, t)) \quad (6)$$

$$\begin{aligned} u(x, t+k) &= u(x, t) + ku_t(x, t) + \frac{k^2}{2}u_{tt}(x, t) + o(k^2) \\ u(x-h, t) &= u(x, t) - hu_x(x, t) + \frac{h^2}{2}u_{xx}(x, t) - \frac{h^3}{6}u_{xxx}(x, t) + o(h^3) \\ u(x-2h, t) &= u(x, t) - 2hu_x(x, t) + 2h^2u_{xx}(x, t) - \frac{4h^3}{3}u_{xxx}(x, t) + o(h^3) \end{aligned} \quad (7)$$

$$u_t = -au_{xx}$$

代入(6)可得:

$$\tau(x, t) = -aku_x + \frac{k^2}{2}a^2u_{xx} + \frac{ak}{2h}(2hu_x - \frac{2}{3}h^3u_{xxx} + o(h^3)) - \frac{a^2k^2}{2h^2}(h^2u_{xx} - h^3u_{xxx} + o(h^3)) = o(h^2 + k^2)$$

(12.89b) 可类似得到。

Exercise 12.79

(12.89a) 用 Fourier 变换:

$$\begin{aligned} U_j^{n+1} &= \frac{1}{\sqrt{2\pi}} \int_{-\frac{\pi}{h}}^{\frac{\pi}{h}} [e^{ijh\xi} - \frac{\mu}{2}(3e^{ijh\xi} - 4e^{i(j-1)h\xi} + e^{i(j+1)h\xi}) + \frac{\mu^2}{2}(e^{ijh\xi} - 2e^{i(j-1)h\xi} + e^{i(j+1)h\xi})] \hat{U}_j^n(\xi) d\xi \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\frac{\pi}{h}}^{\frac{\pi}{h}} e^{ih\xi} \hat{U}_j^{n+1}(\xi) d\xi \end{aligned} \quad (8)$$

如 Exercise 12.49 可得

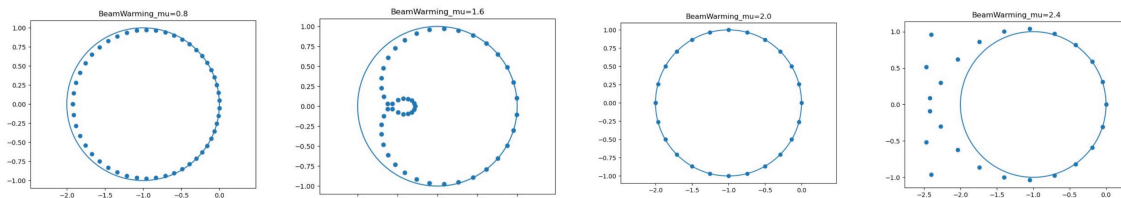
$$\begin{aligned} \hat{U}^{n+1}(\xi) &= [(1 - \frac{3}{2}\mu + \frac{\mu^2}{2}) + (2\mu - \mu^2)e^{-ih\xi} + (\frac{\mu^2 - \mu}{2}e^{-2ih\xi})] \hat{U}^n(\xi) \\ &= e^{-ih\xi} [(1 - 2\mu + \mu^2) \cos h\xi + (1 - \mu)i \sin h\xi + (2\mu - \mu^2)] \hat{U}^n(\xi) \\ &= e^{-ih\xi} [(1 - 2\mu + \mu^2)(-2 \sin^2 \frac{h\xi}{2}) + (1 - \mu)i \sin h\xi + 1] \hat{U}^n(\xi) \end{aligned} \quad (9)$$

要使

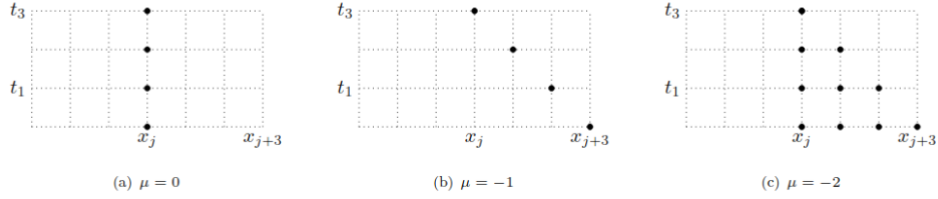
$$\begin{aligned} |e^{-ih\xi} [(1 - 2\mu + \mu^2)(-2 \sin^2 \frac{h\xi}{2}) + (1 - \mu)i \sin h\xi + 1]| &\leq 1 \\ |(1 - 2\mu + \mu^2)(-2 \sin^2 \frac{h\xi}{2}) + (1 - \mu)i \sin h\xi + 1|^2 &\leq 1 \\ 4\mu(2 - \mu)(1 - \mu^2) \sin^4(\frac{h\xi}{2}) &\geq 0 \end{aligned} \quad (10)$$

解得 $\mu \in [0, 2]$.

(12.89b) 类似可得。



Exercise 12.82



Exercise 12.83



Exercise 12.92

Leapfrog method: $\frac{v(x, t+k) - v(x, t-k)}{2k} + a \frac{v(x+h, t) - v(x-h, t)}{2h} = 0$, 在 (x, t) 处 Taylor 展开并且假设 $h=O(h)$, 可得

$$v_t + av_x = -\frac{1}{6}(k^2 v_{ttt} + ah^2 v_{xxx}) - \frac{1}{120}(k^4 v_{ttttt} + ah^4 v_{xxxxx}) + O(k^5) \quad (11)$$

(11)对 t 求导:

$$v_{tt} = -av_{xt} - \frac{k^2}{6} v_{tttt} - \frac{ah^2}{6} v_{xxxt} + O(k^3) \quad (12)$$

(11)对 x 求导:

$$v_{xt} = -av_{xx} - \frac{k^2}{6} v_{xttt} - \frac{ah^2}{6} v_{xxxx} + O(k^3) \quad (13)$$

(13) $\times a$ - (12):

$$v_{tt} = a^2 v_{xx} - \frac{k^2}{6} (v_{tttt} - av_{xttt}) - \frac{ah^2}{6} (v_{xxxt} - av_{xxxx}) + O(k^3) \quad (14)$$

(14)对 t 求导:

$$v_{ttt} = a^2 v_{xxt} - \frac{k^2}{6} (v_{ttttt} - av_{xtttt}) - \frac{ah^2}{6} (v_{xxxtt} - av_{xxxxt}) + O(k^3) \quad (15)$$

(13)对 x 求导:

$$v_{xxt} = -av_{xxx} - \frac{k^2}{6} v_{xttt} - \frac{ah^2}{6} v_{xxxx} + O(k^3) \quad (16)$$

(16) $\times a^2$ + (15):

$$v_{ttt} = -a^3 v_{xxx} - \frac{k^2}{6} (v_{ttttt} - av_{xtttt} + a^2 v_{xxttt}) - \frac{ah^2}{6} (v_{xxxtt} - av_{xxxxt} + a^2 v_{xxxxx}) + O(k^3) \quad (17)$$

(13)对 x 求两次导:

$$v_{xxxxt} = -av_{xxxxx} + O(k^2) \quad (18)$$

(12)对 x 求三次导, 代入(18):

$$v_{xxxxt} = a^2 v_{xxxxx} + O(k) \quad (19)$$

(14)对 x 求两次导, 代入(18):

$$v_{xxttt} = -a^3 v_{xxxxx} + O(k) \quad (20)$$

(16)对 x, t 各求一次导, 代入(19)

$$v_{xtttt} = a^4 v_{xxxxx} + O(k) \quad (21)$$

(17)对 t 求两次导, 代入(19):

$$v_{ttttt} = -a^5 v_{xxxxx} + O(k) \quad (22)$$

将上式代入(11):

$$\begin{aligned} v_t + av_x + \frac{ak^2}{6}(1 - \mu^2)v_{xxx} + O(k^3) &= \epsilon_f v_{xxxxx} \\ v_t + av_x + \frac{ak^2}{6}(1 - \mu^2)v_{xxx} &= 0 \end{aligned} \quad (23)$$

Lax-Wendroff 同理。

Exercise 12.93

$$u_t = -au_x \text{ 由 (12.89a): } \frac{v(x, t+k) - v(x, t)}{k} + a \frac{3v(x, t) - 4v(x-h, t) + v(x-2h, t)}{2h} = ka^2 \frac{v(x, t) - 2v(x-h, t) + v(x-2h, t)}{2h^2}.$$

$$v_t + av_{xx} = \frac{k}{2}(a^2 v_{xx} - v_{tt}) - \frac{k^2}{6}v_{ttt} + \left(\frac{ah^2}{3} - \frac{kha^2}{2}\right)v_{xxx} + O(k^3) \quad (24)$$

(24)对 t 求导:

$$v_{tt} = -av_{xt} - \frac{k}{2}v_{ttt} + \frac{ka^2}{2}v_{xxt} + O(k^2) \quad (25)$$

(24)对 x 求导:

$$v_{xt} = -av_{xx} - \frac{k}{2}v_{xtt} + \frac{ka^2}{2}v_{xxx} + O(k^2) \quad (26)$$

(25)对 x 求导:

$$v_{xtt} = -av_{xxx} + O(k) \quad (27)$$

(25)对 x 求导代入(27):

$$v_{xtt} = a^2 v_{xxx} + O(k) \quad (28)$$

(25)对 t 求导, 代入(28):

$$v_{ttt} = -a^3 v_{xxx} + O(k) \quad (29)$$

(25) - a×(26):

$$v_{tt} = a^2 v_{xx} + O(k^2) \quad (30)$$

代入(11)忽略高阶项有

$$v_t + av_x + \frac{ah^2}{6}(-2 + 3\mu - \mu^2)v_{xxx} = 0.$$

由 $C_p(\xi), C_g(\xi)$ 的定义可知

$$\begin{aligned} C_p(\xi) &= a + \frac{ah^2}{6}(\mu - 1)(\mu - 2)\xi^2 \\ C_g(\xi) &= a + \frac{ah^2}{2}(\mu - 1)(\mu - 2)\xi^2 \end{aligned} \quad (31)$$

因为在 question(e) 中, $\mu = 0.8 < 1 < 2$, 所以数值解的 $C_p(\xi)$ 和 $C_g(\xi)$ 比真解大, 所以数值解的波动有前移。

Exercise 12.94

当 $\mu = 1$ 时, 两个方法修正方程的 $\frac{ah^2}{6}v_{xxx}(1 - \mu)$ 项被消除了, 使得数值解更加精确。

Exercise 12.96

Lax-Friedrichs method:

$$U_j^{n+1} = \frac{1}{2}(U_{j+1}^n - U_{j-1}^n) - \frac{\mu}{2}(U_{j+1}^n - U_{j-1}^n) \quad (32)$$

Fouier 变换:

$$\begin{aligned} U_j^{n+1} &= \frac{1}{\sqrt{2\pi}} \int_{-\frac{\pi}{h}}^{\frac{\pi}{h}} \left[\frac{1}{2}(e^{i(j+1)h\xi} - e^{i(j-1)h\xi}) - \frac{\mu}{2}(e^{i(j+1)h\xi} - e^{i(j-1)h\xi}) \right] \hat{U}_j^n(\xi) d\xi \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\frac{\pi}{h}}^{\frac{\pi}{h}} e^{ijh\xi} \hat{U}_j^{n+1}(\xi) d\xi \end{aligned} \quad (33)$$

$$\begin{aligned} \hat{U}^{n+1}(\xi) &= \left(\frac{e^{ih\xi} - e^{-ih\xi}}{2} - \mu \frac{e^{ih\xi} - e^{-ih\xi}}{2} \right) \hat{U}^n(\xi) \\ &= [\cos h\xi - i\mu \sin h\xi] \hat{U}^n(\xi) \end{aligned} \quad (34)$$

即 $g(h\xi) = \cos h\xi - i\mu \sin h\xi$

$$\begin{aligned} |g(\xi h)| < 1 &\Leftrightarrow 1 + (\mu^2 - 1) \sin^2 h\xi \leq 1 \\ &\Leftrightarrow |\mu| \leq 1 \end{aligned} \quad (35)$$

Exercise 12.97

Lax-Wendroff method:

$$U_j^{n+1} = U_j^n - \frac{\mu}{2}(U_{j+1}^n - U_{j-1}^n) + \frac{\mu^2}{2}(U_{j+1}^n - U_j^n + U_{j-1}^n) \quad (36)$$

$$\begin{aligned} U_j^{n+1} &= \frac{1}{\sqrt{2\pi}} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left[e^{ij\xi h} - \frac{\mu}{2}(e^{i(j+1)\xi h} - e^{i(j-1)\xi h}) + \frac{\mu^2}{2}(e^{i(j+1)\xi h} - e^{ij\xi h} + e^{i(j-1)\xi h}) \right] \hat{U}_j^n(\xi) d\xi \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^{ij\xi h} \hat{U}_j^{n+1}(\xi) d\xi \end{aligned} \quad (37)$$

同上可知

$$\hat{U}^{n+1}(\xi) = \left[1 - 2\mu^2 \sin^2 \frac{\xi h}{2} - i\mu \sin \xi h \right] \hat{U}_j^n(\xi) \quad (38)$$

即 $g(h\xi) = 1 - 2\mu^2 \sin^2 \frac{\xi h}{2} - i\mu \sin \xi h$

$$\begin{aligned} |g(\xi h)| < 1 &\Leftrightarrow |g(\xi h)|^2 = 1 + \mu^2(\mu^2 - 1) \sin^4 \frac{\xi h}{2} \leq 1 \\ &\Leftrightarrow |\mu| \leq 1 \end{aligned} \quad (39)$$